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# <Pulkit Saxena>                                <25.November.2023>
# <Assignment 6>
# Description: This assignment is based on the game known as candy realm. In this
assignment, we have created different functions with different arguments which help
us to create the game in an efficient manner. In the game, the user can choose
whether he wants to play the game, read the instructions or quit the game. He can
also determine the no. of players who are going to play this game and the no. of
copies required for each card.

import random

def printTitleMaterial():
    """Prints the title material for the game, including the student's name, class,
    and section number.
    """
    print("Candy Realm!")
    print()
    print("By: Pulkit Saxena")
    print("[COM S 127 J]")
    print()

def Deck(numcopies):
    """Initializes and shuffles the deck of Candy Realm cards."""
    cards = ["M", "R", "B", "C", "G", "Y"]
    deck = cards * numcopies
    random.shuffle(deck)
    return deck

def printBoard(playerpositions):
    """Prints the Candy Realm game board."""
    print("-" * 65)
    print("1\n2\n3\n4\nSTART M R B M C G B Y C B G Y G C Y R M R GOAL!")
    print("CARDS", " ".join(playerpositions))
    print()

def drawCard(deck):
    """Draws a card from the deck."""
    if not deck:
        print("The deck is empty. Reshuffling...")
        return Deck()
    return deck.pop(0)

def Game(numplayers, num_copies):
    """Plays the Candy Realm game."""
    print(f"Welcome to Candy Realm! Playing with {numplayers} players and
    {num_copies} copies for each card.")

    # Ensure the number of copies is between 1 and 5
    num_copies = max(1, min(num_copies, 5))

    deck = Deck(num_copies)
    player_positions = ["M"] * numplayers

    while True:
        printBoard(player_positions)

        for i in range(numplayers):
            player_input = input(f"Player {i + 1}: Would you like to [d]raw a
            {player_positions[i]} card, ")

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        "[s]huffle the deck, or [q]uit?: ")

    if player_input == 'd':
        card_drawn = drawCard(deck)
        print(f"Player {i + 1} has drawn: {card_drawn}")
        move = deck.count(card_drawn)
        player_positions[i] = card_drawn
        print(f"Player {i + 1} moves forward {move} spaces!")

        # Display player's position in the deck
        print(f"Player {i + 1}'s position in the deck:
{deck.index(card_drawn) + 1}")

        input("Press [ENTER] To Continue...")

        # Check if the player has reached or exceeded the goal
        if player_positions[i] == "GOAL":
            print(f"Player {i + 1} is the winner!")
            return

    elif player_input == 's':
        random.shuffle(deck)
        print(f"Player {i + 1} has shuffled the deck!")
        input("Press [ENTER] To Continue...")
    elif player_input == 'q':
        print("Thanks for playing! Goodbye!")
        return

def Instructions():
    """Displays game instructions."""
    print("Instructions for Candy Realm:")
    print("1. The goal of the game is to reach the 'GOAL' space first.")
    print("2. Players start on the 'START' space and move forward by drawing
cards.")
    print("3. Each player can draw a card, shuffle the deck, or quit during their
turn.")
    print("4. The card drawn determines how many spaces a player moves forward.")
    print("5. The first player to reach the 'GOAL' space wins.")
    print("6. Good luck and have fun!")

def main():
    """Main function to start the game."""
    printTitleMaterial()

    while True:
        print("MAIN MENU: [p]lay game, [i]nstructions, or [q]uit?: ")
        choice = input().lower()

        if choice == 'p':
            num_players = int(input("Enter the number of players (2-4): "))
            num_copies = int(input("Enter the number of copies for each card (1-5):
"))

            # Ensure the number of copies is between 1 and 5
            if 2 <= num_players <= 4 and 1 <= num_copies <= 5:
                Game(num_players, num_copies)
            else:
                print("Invalid input. Please enter valid values.")
        elif choice == 'i':

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        Instructions()
    elif choice == 'q':
        print("Goodbye! Thanks for playing.")
        break
    else:
        print("Invalid choice. Please enter 'p', 'i', or 'q'.")

if __name__ == "__main__":
    main()
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